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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations
December 2025
EE2312-1 – MICRO AND NANO SCALE SENSORS AND TRANSDUCERS
(For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. BaSrTiO deposited using electrophoretic technique for
 - A) Ultra sensor
 - B) capacitive pressure sensor
 - C) Acceleration sensor
 - D) Moisture sensor
2. 555 timer in capacitive pressure sensor converts capacitance to _____
 - A) Frequency
 - B) voltage
 - C) current
 - D) oscillation
3. Inductive pressure sensor has zero inductance when the core has ____ air permeability
 - A) full
 - B) 30%
 - C) zero
 - D) 50%
4. Inductive pressure initial _____ is applied on spring
 - A) voltage
 - B) current
 - C) pressure
 - D) flux
5. _____ pressure sensor is used for vibrations
 - A) capacitance
 - B) Ultra high sensitivity
 - C) inductance
 - D) displacement
6. _____ have a property of superhydrophobic
 - A) MOSFET
 - B) thermistor
 - C) BJT
 - D) CNT's
7. Ultrahigh Sensitivity, Wide Dynamic Range Sensors has property of linear upto
 - A) step
 - B) ramp
 - C) 100g
 - D) 0
8. CCD can be easily _____ at a Nyquist rate for high-bandwidth music
 - A) clipped
 - B) sampled
 - C) clamped
 - D) filtered
9. A Lab on a Chip is essentially composed of three sections _____
 - A) actuation, sensing, and electronic interface circuitry
 - B) actuation, sensing, and Image processing
 - C) actuation, image processing, and electronic interface circuitry
 - D) actuation, sensing, and clipping
10. in lab on chip the electronic interface circuitry finally performs traditional signal processing functions such as _____
 - A) amplification and noise reduction
 - B) coarse partition
 - C) clipping and clamping
 - D) image processing and filtering
11. Beta particles emitted from the decay of _____ have a maximum energy of 18.6 keVs
 - A) tin
 - B) Biso3
 - C) Basrtio3
 - D) tritium

12. solid-state magnetic field sensor has _____ power consumption
 A) maximum B) moderate
 C) minimum D) Zero
13. electron travels inside the semiconductor lattice, it loses kinetic energy _____
 A) exponentially B) step
 C) linearly D) nonlinear
14. _____ technology is useful for battery power consumer
 A) flux creation B) Electromagnetics
 C) electrostatic D) electrolytic
15. The CCD used in the present prototype is a low resolution _____
 A) (640 × 480 pixels) B) (640 × 48 pixels)
 C) (6400 × 480 pixels) D) (60 × 480 pixels)
16. The total harmonic distortion (THD) was tested near the _____ range of the dynamic range of the microphone
 A) Low B) moderate
 C) noise D) peak
17. An _____ source is attached to the wing of the aircraft
 A) Wheel B) beta particle
 C) alpha-particle D) sensor
18. a particles strike the detector, they deposit their _____ charges on the gate of the MOSFET
 A) moisture B) electron
 C) positive D) Vdd
19. The amount of radioactive isotope is _____ than conventional in smoke detector
 A) more B) less
 C) neutral D) active
20. A layer of water will _____ the alpha particles in the sensor
 A) activate B) stop
 C) fasten D) moderate

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List the limitations of conventional capacitive pressure sensor.	4	L*1	1	1,2
	b) Explain the construction of the Inductive pressure sensor.	6	L2	1	1,2
	c) With neat sketch explain working principle of smoke detector.	6	L3	2	1,2,7
2.	a) Derive the expression for pressure P acting on the mercury drop as a function of capacitance.	6	L2	1	1,2
	b) Explain with neat sketch the working of ultra-sensitivity pressure sensor.	6	L2	2	1,2
	c) List the applications of CO based gas sensor.	4	L1	2	1,2
3.	a) Explain the following characteristic of capacitor pressure sensor i) pressure Vs displacement (ii) pressure vs capacitance.	4	L2	1	1,2
	b) List the application of moisture sensor sensor.	4	L1	2	1,2
	c) Write a short note on measurement of capacitance constant in acceleration sensor with an example.	8	L3	2	1,2
Unit – II					
4.	a) Explain mechanical structure integration of microphone assembly.	6	L2	3	2,7
	b) With an example explain the application of lab on chip sensors.	6	L3	4	2,7
	c) Discuss the impact of temperature on the response of the solid state magnetic field sensor.	4	L1	3	2,7

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|-------|---|---|----|---|-------|
| 5. a) | Explain the procedure followed to measure the total harmonic distortion in microphone. | 6 | L3 | 3 | 2,7 |
| b) | With a neat sketch explain working principle of magnetic field sensor. | 6 | L1 | 4 | 2,7 |
| c) | Explain the advantage of solid state magnetic field sensor over convention magnetic field sensor. | 4 | L2 | 4 | 2,7 |
| 6. a) | Explain the frequency response characteristics and microphone's self-noise. | 6 | L2 | 3 | 2,7 |
| b) | Draw schematic diagram of (i) CMOS on-chip sensor for measuring the dielectric constant of organic chemicals,
(ii) Porous silicon based sensor for chemical gas vapor detection. | 6 | L2 | 4 | 3,4,7 |
| c) | List the advantages of magnetic field sensor. | 4 | L1 | 4 | 4,7 |

Unit – III

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|-------|--|---|----|---|-----|
| 7. a) | Compare the difference between microfluid and micro actuator devices. | 6 | L2 | 5 | 1,2 |
| b) | With a neat sketch explain the fundamental principle of operation the new icing detector. | 6 | L3 | 5 | 2 |
| c) | List the advantages of micro fluid sensor in aircraft. | 4 | L1 | 5 | 2 |
| 8. a) | With a neat sketch explain the construction and working principle of microfluid actuators. | 8 | L2 | 5 | 3,7 |
| b) | Explain the construction and working principle of microactuators. | 8 | L3 | 5 | 4,7 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
